

PROJECT TITLE

VisEvi.

Disaster signals, verified from orbit.

A sensor or weather signal raises the alarm. VisEvi captures a fresh satellite view of the site, compares it with the same season a year earlier through Cloudinary, confirms or questions the alert, and tells the responsible person.

19

monitored sites

117

satellite and field captures

95%

passing verification

Team name

[Your Team Name]

Members

[Member 1, role] · [Member 2, role] · [Member 3, role] · [Member 4, role]

Tracks chosen

Sustainability

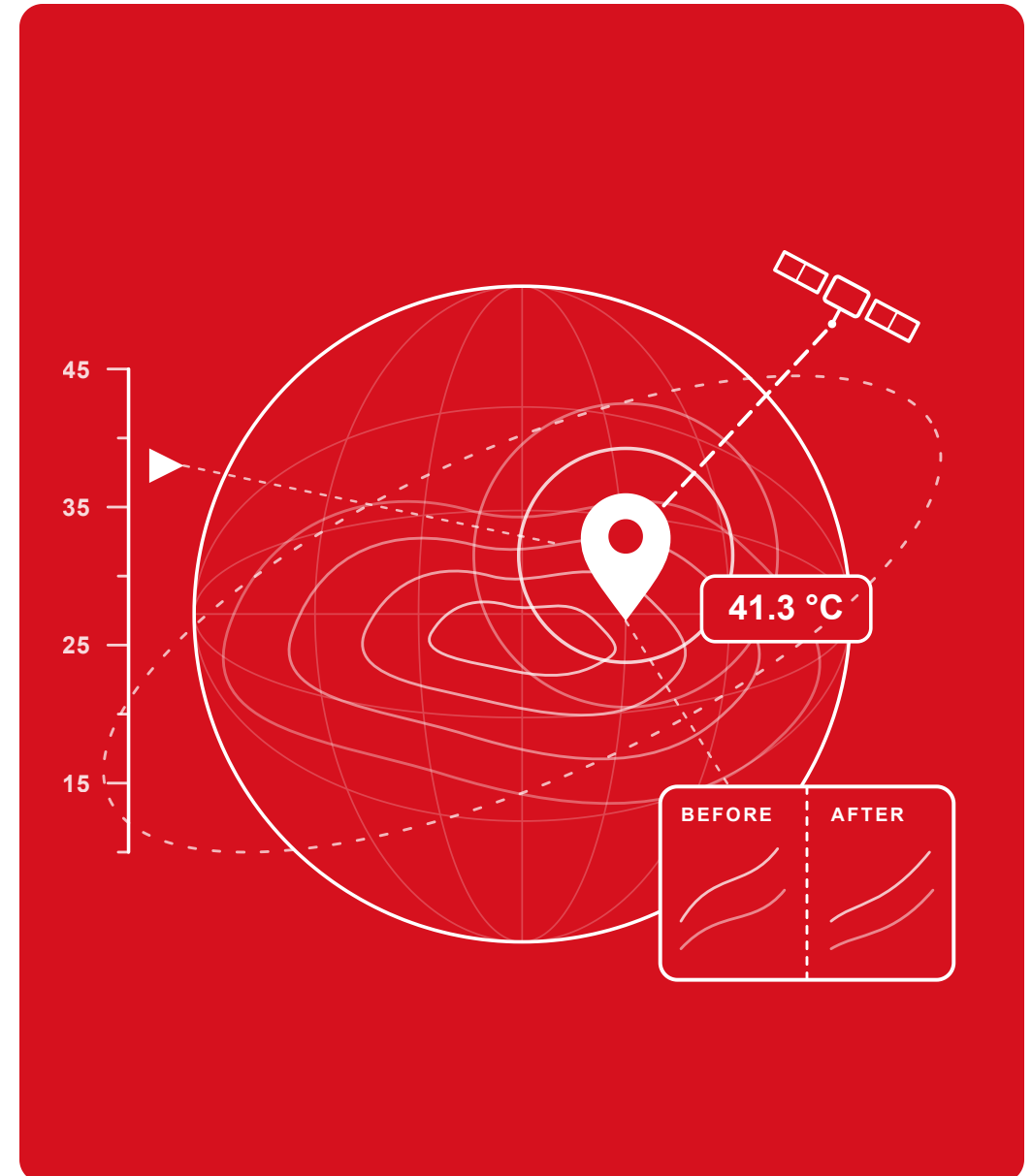
AI / ML

Internet of Things

Healthcare

Web3 Development

Open Innovation



PROBLEM STATEMENT AND TARGET AUDIENCE

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A warning is not proof, and proof rarely reaches the right person in time.

The problem

- **Signals and evidence are disconnected.** A forecast or sensor produces a number, but nobody can quickly see, verify and document what is happening on the ground.
- **Field evidence is manual and easy to fake.** Photos are sorted by hand, reused across reports, and hard to trace to their source.
- **Warning coverage is incomplete.** Half of the world's countries still lack multi-hazard early warning systems; the UN goal is coverage for everyone by 2027.
- **Even impact figures are contested.** Press reports of India's national surveillance data give 56 confirmed heatstroke deaths for March to May 2024 and 14 for March to June 2024.

7,348

recorded
disasters, 2000 to
2019

1.23 M

lives lost

4.2 B

people affected

US\$2.97 T

economic losses

Climate-related disasters rose from 3,656 (1980 to 1999) to 6,681 (2000 to 2019). Sources: UNDRR and CRED, 2020; UNDRR and WMO, 2022 (see References).

Who faces it

B2G: PRIMARY

State and district disaster management authorities, municipal corporations, control-room officers. Typically professionals in their late 20s to 50s.

B2B

Parametric and crop insurers verifying events, utilities and infrastructure operators, agri-cooperatives.

NGOS AND SUSTAINABILITY TEAMS

Humanitarian programme teams, CSR and ESG auditors who must prove what happened, where and when.

INDIRECT BENEFICIARIES

Communities in heat-, flood- and cyclone-prone regions who get a faster, better-founded warning. No B2C product at this stage.

OUR UNIQUE SOLUTION

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Signal, satellite proof, verdict, notification: one closed loop.

- 1

Signal watch
Live temperature and coastal waves at every monitored site, judged against ten years of local history plus fixed limits. Field sensors post to a webhook.
- 2

Automatic satellite capture
An alert pulls a fresh NASA Worldview snapshot of the site, screens out cloud and no-data days, and sends it through Cloudinary (enhance, AI Vision).
- 3

Before / after verdict
Compared with the same season a year earlier; vegetation, water-like area and cloud are measured. Result: confirmed, not confirmed or inconclusive.
- 4

Notify the right person
Responsible contacts get email or free ntfy push, with the evidence and a full audit log of every message.
- 5

Risk index with confidence
0 to 100 from six sourced factors (NASA hazard maps, live, forecast, incidents, nearby events, imagery), plus a separate confidence score.
- 6

Trust layer
Satellite images are re-requested from NASA to prove provenance; duplicates, wrong content and false dates are flagged; every asset traces to its Cloudinary ID and transformations.

Why it is different

EXISTING ALTERNATIVE	WHAT IT DOES	WHAT IS MISSING
Weather and warning apps	Publish a forecast or alert	No proof from the ground, no per-site follow-up
Satellite viewers	Show imagery on demand	Manual, no trigger, no verdict, no notification
Global event feeds	List fires, storms, floods	Not checked at your site; no owner told
Media managers	Store and transform photos	No live signal, no verification

Novelty

- **Sensor-triggered, satellite-verified alerts** in one product.
- **Honest by design:** "inconclusive" is a real answer; a metric that could not be calibrated was dropped, not shown.
- **Cloudinary beyond storage:** pre-processing, AI Vision, provenance, and a one-URL before/after board with the reading stamped on it.

TECH STACK AND ARCHITECTURE

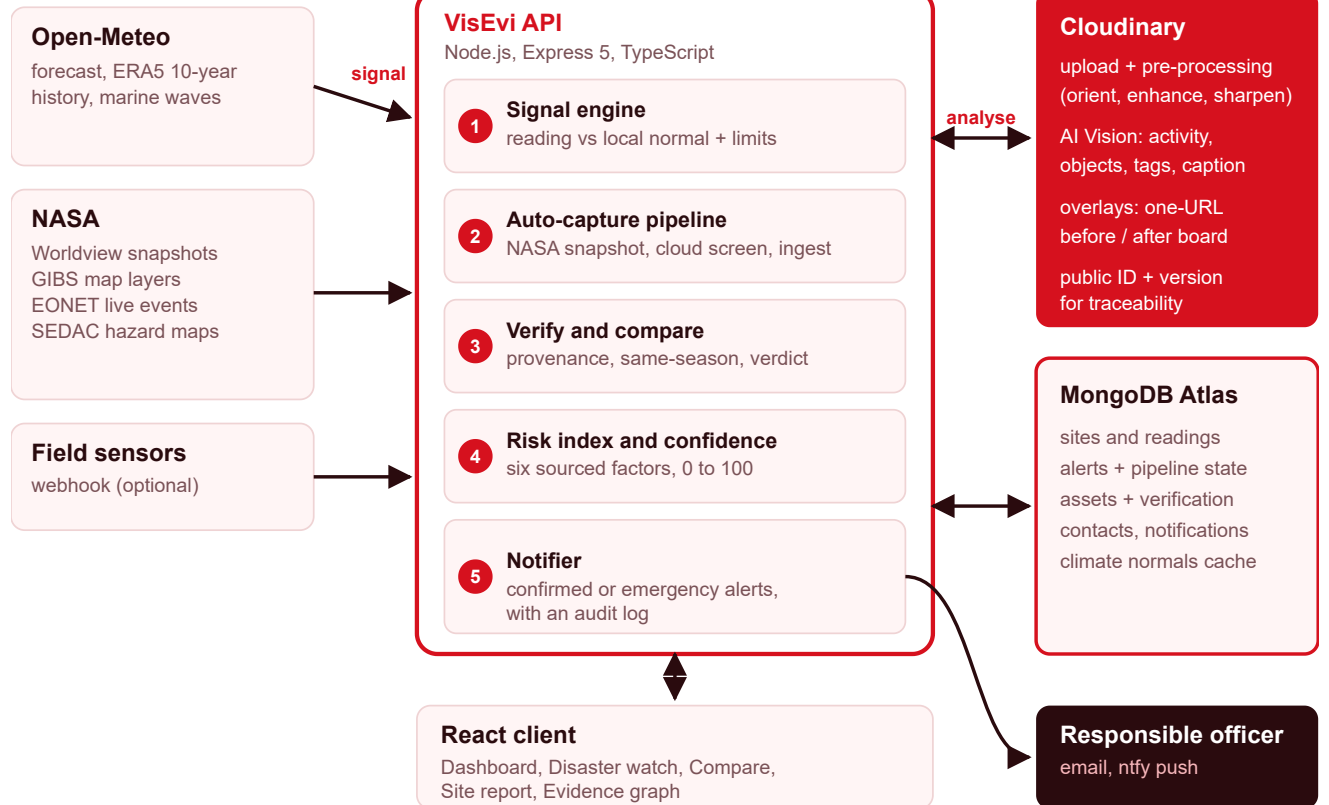
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Free public data in, verified alerts out.

Tech stack

Frontend	React 19, Vite, TypeScript, Tailwind CSS 4, React Router, Leaflet, hand-built SVG charts
Backend	Node.js, Express 5, TypeScript, sharp (image analysis), Nodemailer, Multer
Database	MongoDB Atlas: sites, readings, alerts, assets, contacts, notifications
Media + AI	Cloudinary: upload, auto-orient / enhance / sharpen, AI Vision, overlay transformations
Data APIs	Open-Meteo (forecast, ERA5 history, marine); NASA GIBS, Worldview Snapshots, EONET, SEDAC hazard maps; OpenStreetMap and Nominatim
Alerts out	SMTP email, ntfy.sh push
Deploy	Vercel (client), Render (API), Atlas

PUBLIC DATA + SENSORS



Flow: **1** reading breaches a limit or the local normal, **2** alert triggers a NASA capture, **3** Cloudinary analyses it, and it is compared and given a verdict, **4** risk is recomputed, **5** the responsible officer is told.

Built, running, and honest about its limits.

Feasibility: it already runs

- **Working prototype:** 19 monitored sites, 117 captures across 23 projects (95% verified), 22 alerts, including a replayed May 2024 heat wave (Barmer 48.0 °C).
- **Zero paid data:** every source is free and keyless; only Cloudinary and MongoDB accounts are needed.
- **Tested end to end** with real NASA and Cloudinary calls. The 15-minute poll raised and captured a live alert without anyone pressing a button.
- **Team fit:** full-stack TypeScript, geospatial data, Cloudinary; built in incremental milestones inside the hackathon window.

Showstoppers and mitigation

RISK	MITIGATION
Satellite resolution is about 250 m per pixel	"Inconclusive" is a first-class verdict; roadmap: Sentinel-2 at 10 m
Cloud cover and no-data days	Automatic screening, retry over 10 days, reuse a recent capture
Free-API terms and limits	Caching, provider swap (Google Weather optional), paid plans when commercial
Cloudinary quota	Capture only when an alert fires; reuse within 8 days
False alarms , unvalidated weights	Anomaly must also be stressful; confidence score; back-test against EM-DAT next

Feasibility, showstoppers, USP and business model

USP

One product closes the loop from a live signal to a satellite-verified, notified alert, with a transparent risk score and a confidence value that says how far to trust it.

- Confirms or questions every alert with imagery.
- Provenance-checked evidence, not just organised evidence.
- Free public data keeps the cost of each site low.

Business model

Customers	State and district disaster authorities (B2G); insurers and utilities (B2B); NGOs and CSR or ESG auditors
Subscription	Per monitored site per month, tiered by site count and check frequency
Verified-event API	Per event or claim check for insurers, via webhook
Services	Onboarding: thresholds, contacts and local-language setup
Free tier	NGO and community use, built on open data
Sustainability	Costs are event-driven (one AI analysis per capture, only on alerts), so margin grows with sites, not with time

VisEvi

REFERENCES

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Where the numbers and methods come from.

1 Research papers and journals

- Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146, 1999-2049.
- Dilley, M. et al. (2005). *Natural Disaster Hotspots: A Global Risk Analysis*. World Bank.
- Justice, C. O. et al. (1998). MODIS: land remote sensing for global change research. *IEEE TGRS* 36(4).
- Woebbecke, D. M. et al. (1995). Color indices for weed identification. *Trans. ASAE* 38(1) (excess-green index).

2 Official documentation

- Cloudinary documentation: upload, transformations, layers, AI Vision. cloudinary.com/documentation
- NASA GIBS API docs. nasa-gibs.github.io/gibs-api-docs
- NASA EONET API v3. eonet.gsfc.nasa.gov/docs/v3
- Open-Meteo API docs. open-meteo.com/en/docs

3 Open-source libraries

- React, Vite, Tailwind CSS, React Router, Leaflet (client)
- Express, MongoDB Node driver, sharp, Nodemailer, Multer, Cloudinary SDK (server)

4 APIs and developer documentation

- Open-Meteo forecast, archive and marine APIs; NASA Worldview Snapshots and GIBS WMTS; NASA EONET
- OpenStreetMap tiles and Nominatim geocoding; [ntfy.sh push](https://docs.ntfy.sh) (docs.ntfy.sh)

5 Datasets and data sources

- ERA5 hourly temperature via Open-Meteo archive (ten-year local climate normals)
- NASA SEDAC Natural Disaster Hotspots: drought, flood, cyclone, landslide layers, served by GIBS
- MODIS Terra corrected reflectance and land surface temperature; SMAP soil moisture; GPM IMERG rainfall
- EM-DAT (CRED) disaster database, as summarised by UNDRR

6 Industry reports and case studies

- UNDRR and CRED (2020). *The Human Cost of Disasters 2000-2019*. undrr.org/publication/human-cost-disasters-2000-2019
- UNDRR and WMO (2022). *Global Status of Multi-Hazard Early Warning Systems*.
- UN Early Warnings for All initiative. wmo.int/activities/early-warnings-all

7 Other online resources

- Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR)
- India Meteorological Department heat wave criteria (basis of the fixed thresholds)
- Krawetz, N. (2013). *Kind of Like That* (difference-hash image fingerprinting)
- Press reports of NCDC heatstroke data: Business Today, 1 Jun 2024 (56 deaths, March to May); Deccan Herald, 28 Jul 2025 (14 deaths, March to June)